

AEGIS: One-Query Adversarial Diagnostics over the GNN Vulnerability Spectrum

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Abstract—Graph neural networks in safety-critical settings (drug interaction, fraud detection, power-grid contingency screening) are vulnerable to adversarial graph perturbations, yet none derives, from a single analysis, the three diagnostics a practitioner needs: a maximally sensitive first-order direction, per-edge equilibrium-sensitivity rankings, and per-node sensitivity radii. AEGIS produces all three from one object, the *constrained sensitivity matrix* S_c , computed matrix-free via a Neumann-series resolvent and randomized SVD that scale to $N=7,650$ on a single GPU. For the spectral-norm-constrained IGNN subclass we prove a closed-form three-regime characterisation with critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ that trained models satisfy with $2\text{--}4\times$ margin. S_c extends to continuous-edge-weight message passing (GCN, SAGE, GIN, APPNP, IGNN, edge-weighted GAT^T): the continuous-to-discrete bridge is positive in 29/33 architecture–dataset cells ($p<10^{-5}$) and reaches $\tau = +0.996\pm0.002$ (10 seeds) against brute-force N-1 on full-graph Amazon Photo ($N=7,650$), while the one-query SVD direction matches 50-step PGD attackers. Top- v_{ij} masking cuts $\sigma_1(S_c)$ -damage $42\pm8\%$ (vs. 11% random) and survives adaptive recomputation; on IEEE power flow S_c recovers N-1 rankings on case14–118 (P@10=0.66–0.81), competitive with industry LODF.

Index Terms—Graph neural networks, adversarial robustness, constrained sensitivity, implicit graph neural networks, randomized SVD, power-flow contingency screening

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection via perturbed transaction edges [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Structural attacks [1], [4], [5], [6] give per-edge surrogate gradients but no continuous direction or per-node budgets; smoothing [7], [8], [9], [10] and IBP-style certifiers [11], [12] give per-node certificates but no edge rankings or direction; robust-architecture defenses [13], [14], [15] harden models without surfacing per-edge vulnerability; sober-look critiques [16], [17] flag evaluation pitfalls. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) produces all three from one analytical object, S_c (section IV)—“one query” meaning a single matrix-free S_c construction (one randomized SVD over the equilibrium resolvent), not three separate analyses. The closed-form regime characterisation (theorem 1) is established for contractive implicit GNNs and S_c extends to any GNN with continuous edge-weight-modulated message

passing (section V-F); the threat model targets edge-deletion and weight-perturbation (section II).

Contributions. (1) **Constrained sensitivity operator.** S_c specialises equilibrium IFT sensitivity [18], [19] to *structural* edge perturbations via the projection P_c ; one matrix-free query reads the SVD-optimal direction, per-edge rankings, and per-node radii (dense-matching σ_1 to 0.03% at $N=200$; scaling to $N=7,650$)—no prior object yields all three at once. (2) **Phase-transition theory.** A worst-case three-regime characterisation with closed-form critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ for the spectral-norm-constrained IGNN subclass, which trained IGNNs satisfy with $2\text{--}4\times$ margin across our κ_{max} sweep, giving practitioners a provable structural-perturbation safety threshold ($\sim 6\%$ accuracy cost; theorem 1 and section V-D). (3) **Empirical evaluation.** 9 datasets, 7 architectures, 4 domains (330 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD [5], AGNNCert [11], and LODF (table IV), continuous-to-discrete transfer (fig. 7), and a power-grid N-1 case study (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes and adjacency $A \in \mathbb{R}^{N \times N}$, node features $X \in \mathbb{R}^{N \times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i -th singular value, and $\text{vec}(\cdot)$ the column-major vectorization. The sensitivity objects introduced below are: the unconstrained $S \in \mathbb{R}^{Nd \times N^2}$, constrained $S_c \in \mathbb{R}^{Nd \times |E|}$, unrolled $S_K \in \mathbb{R}^{Nd \times N^2}$ (theorem 8), and per-node block S_v ; the contractivity $\kappa = \|J_z\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\text{crit}}$, and per-node radius r_v .

IGNN and IFT. A DEQ-GNN [20], [21] iterates a weighted F_θ to a fixed point; the IGNN instantiation [22] uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \quad (1)$$

with $W \in \mathbb{R}^{d \times d}$ spectral-normalized [23] so $\kappa \leq \|\hat{A}\|_2 \|W\|_2 < 1$, ReLU ϕ , and a learned projection X_{proj} . Training uses implicit differentiation through the fixed point [19], [24] with optional Jacobian regularization [25], and a linear head $f(z^*) = W_{\text{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A , with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p} \Big|_{z^*}, \quad (2)$$

where $\|(I - J_z)^{-1}\|_2 \leq 1/(1-\kappa)$ via the convergent Neumann series, and the pseudospectral index $\eta = \|(I - J_z)^{-1}\|_2 (1 -$

$\rho) \geq 1$ [26] measures the gap to the spectral-radius bound (with $\eta=1$ for normal J_z). The IFT has been used for hyperparameter optimization [27] and implicit differentiation [19], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \|\delta\hat{A}\|_F \leq \varepsilon, \quad (3)$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A} = \delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge-deletion / weight-perturbation; insertion attacks (Nettack-class [1], [4]) require enlarging the basis to $\bar{E} \supseteq E$, a natural follow-up that preserves the S_c construction. Discrete deletions connect to the continuous model through theorem 7. Unlike input-feature perturbation [28], [29], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F / \partial \text{vec}(A)$, motivating S_c in section III.

III. ADVERSARIAL EQUILIBRIUM THEORY

Building on the threat model of section II, we derive the analytical machinery behind the three diagnostics promised in section I: per-edge rankings, the SVD-optimal direction, and per-node radii. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which the three diagnostics follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- (A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [30] (proof below).
- (A2) $\|W\|_2 \leq c$ via spectral normalization.
- (A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training ($\kappa=0.14-0.59$ across all benchmarks in our suite; section V-E). (A2) does not imply (A3) automatically; we report κ alongside $\varepsilon_{\text{crit}}$ so practitioners can audit, and AEGIS continues to return S_c rankings and SVD-optimal attacks if (A3) is not satisfied, with the formal guarantees below then scoped to the contractive regime.

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \quad (4)$$

Then structural perturbations $\delta\hat{A}$ with $\|\delta\hat{A}\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical ($\varepsilon < \varepsilon_{\text{crit}}$): *The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \quad (5)$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} / (1 - \kappa)$.

(b) Critical ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$): *The resolvent obeys the unconditional eigenvalue bound $\|(I - J'_z)^{-1}\|_2 \geq 1 / \min_i |1 - \lambda_i(J'_z)|$, so it blows up precisely when an eigenvalue of J'_z approaches 1—a vanishing spectral gap to 1, not $\|J'_z\|_2 \rightarrow 1$ on its own. The rate is $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ when J'_z is normal with a dominant real-positive eigenvalue (the Perron mode of symmetric $\hat{A}, W$); in general $\varepsilon_{\text{crit}}$ is the norm-based contraction radius that lower-bounds the eigenvalue divergence threshold, the slack reflecting the nonnormality of J'_z (bounded by η , theorem 2).*

(c) Supercritical ($\varepsilon \geq \varepsilon_{\text{crit}}$): *The sufficient contraction certificate no longer applies ($\|J'_z\|_2$ may exceed 1) and the part-(a) first-order guarantees do not apply; the certificate is reported as a sufficient safety boundary, validated empirically by the κ_{max} sweep of section V-D (trained IGNNs hold with 2–4 $\times$ margin).*

Proof. **(a).** Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\|\delta\hat{A}\|$ (the boundary set $\{\hat{A} : z^*(\hat{A}) \text{ lies on a region face}\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels [30] yields a conservative IFT on each region. With $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \|\hat{A}\|_2 \|W\|_2$, applying IFT to $G(z, \hat{A}) = z - F(z, \hat{A}) = 0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \text{vec}(\delta\hat{A}) + O(\|\delta\hat{A}\|^2), \quad (6)$$

with $J_A = \partial F / \partial \text{vec}(A)$ and Neumann $\|(I - J_z)^{-1}\|_2 \leq 1/(1 - \kappa)$ yielding (5). For contractivity preservation, $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $\varepsilon < \varepsilon_{\text{crit}}$ guarantees $\|J'_z\|_2 < 1$.

(b)–(c). For any M , $(I - M)^{-1}$ has eigenvalues $(1 - \lambda_i(M))^{-1}$, hence $\|(I - M)^{-1}\|_2 \geq \rho((I - M)^{-1}) = 1 / \min_i |1 - \lambda_i(M)|$ unconditionally. Along the worst-case direction $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $1 - \|J'_z\|_2 \geq \|W\|_2 (\varepsilon_{\text{crit}} - \varepsilon)$ and, since $\rho \leq \|\cdot\|_2$, $\varepsilon < \varepsilon_{\text{crit}}$ certifies $\|J'_z\|_2 < 1$. The certificate is one-sided: $\|J'_z\|_2 \rightarrow 1$ need not drive $\min_i |1 - \lambda_i| \rightarrow 0$ (e.g. $M = \text{diag}(-s, 0)$ has $\|M\|_2 = s$ yet $\|(I - M)^{-1}\|_2 = 1$). When J'_z is normal with dominant real-positive eigenvalue $\lambda^* = \|J'_z\|_2$ (the Perron mode for symmetric $\hat{A}, W$), $\min_i |1 - \lambda_i| = 1 - \lambda^* = \|W\|_2 (\varepsilon_{\text{crit}} - \varepsilon)$ yields the $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ rate; a negative or complex dominant eigenvalue keeps $\min_i |1 - \lambda_i|$ away from 0 (the same $\text{diag}(-s, 0)$), so $\varepsilon_{\text{crit}}$ only lower-bounds the threshold, slack controlled by η (theorem 2). For $\varepsilon \geq \varepsilon_{\text{crit}}$ Banach contraction fails and the iteration may converge elsewhere, oscillate, or diverge. $\square$

$\varepsilon_{\text{crit}}$ is a closed-form sufficient bound that S_c tightens in practice, 7–14 $\times$ over the unconstrained $\sqrt{\tau}$ bound on 50-node subgraphs (section V-A). The Neumann series requires the operator-norm κ rather than $\rho(J_z)$ [31]; the gap is the pseudospectral index η , bounded by the weight matrix alone:

Observation 2 (Graph-Independent Nonnormality Bound, all-active case). *For $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$ and all activations positive ($\phi' \equiv 1$): $\eta \leq \kappa(V_W)$ where*

$W=V_W D_W V_W^{-1}$ and $\kappa(V_W)=\|V_W\|_2 \|V_W^{-1}\|_2$, independent of $\hat{A}$.

Proof. $J_z=\hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [31]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}})=1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\|(I-M)^{-1}\|_2 \leq \kappa(V)/(1-\rho(M))$ yields $\eta \leq \kappa(V_W)$. $\square$

Remark 3 (Empirical extension to general ReLU patterns). *The proof above requires $\phi' \equiv 1$; for general ReLU patterns, $\eta \in [1.19, 2.47]$ on our suite (section V-E), tracking $\kappa(V_W)$ closely. Graph structure does not amplify nonnormality; the moderate η traces to spectral-norm regularisation of W .*

Proposition 4 (Maximally Sensitive First-Order Structural Perturbation). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S . The maximum first-order equilibrium shift is $\varepsilon \cdot \sigma_1(S)$.*

By the variational characterisation of σ_1 , the direction (theorem 4) is optimal for hidden-state damage within the pseudospectral envelope of theorems 2 and 3, and empirically aligns with classification damage (section V-B). The constrained sensitivity matrix $S_c \in \mathbb{R}^{N \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with edge basis $b_k = (e_i e_j^\top + e_j e_i^\top) / \sqrt{2}$ ($\|b_k\|_F=1$, so $\sigma_1(S_c)$ is consistent with $\|\delta \hat{A}\|_F = \varepsilon$). The $N^2 \rightarrow |E|$ projection P_c is the standard duplication-matrix reduction [32] on the edge-supported symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. The first-order envelope identifies the dangerous direction rather than a global safety margin (table II).

Proposition 5 (Per-Node First-Order Sensitivity Radius). *For a linear head $f(z)=Wz+b$ and a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v}(z_v^*) - f_c(z_v^*) > 0$ to every competitor $c \neq y_v$, the first-order sensitivity radius*

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c) S_v\|_2} \quad (7)$$

preserves the classification of v for any $\|\delta \hat{A}\|_F < r_v$, where S_v are the block-rows of S at node v ; substituting $S_{c,v}$ gives the tighter constrained-threat radius $r_v^{(c)} \geq r_v$. Minimizing over all competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes under perturbation; the cheaper runner-up surrogate $\hat{r}_v = m_v / (\|W_{y_v} - W_{c^}\|_2 \|S_v\|_2)$ ($c^* = \arg \max_{c \neq y_v} f_c$) can be optimistic when a non-runner-up class is more aligned.*

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v \text{vec}(\delta A)$, so for each competitor $|\Delta(f_{y_v} - f_c)| \leq \|(W_{y_v} - W_c) S_v\|_2 \|\delta A\|_F$ by Cauchy-Schwarz; the prediction survives while every margin stays positive, i.e. $\|\delta A\|_F < m_v^{(c)} / \|(W_{y_v} - W_c) S_v\|_2$ for all c , whose minimum is r_v . Intuitively, r_v is the smallest margin divided by two amplification factors (decision-boundary

sensitivity to hidden-state change, and equilibrium sensitivity to structure); high-margin, low-sensitivity nodes get large radii.

Remark 6 (Sensitivity threshold semantics). r_v is a first-order sensitivity threshold, not a probabilistic certificate: it indicates the perturbation scale at which the linearization predicts a classification change for node v , and is locally tight for small ε but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [7] certifies which nodes are guaranteed safe, while AEGIS differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is 0% at $\varepsilon=0.01$ and rises with ε as expected for first-order approximations (section V-B).

A. Continuous-to-Discrete Transfer

Proposition 7 (Continuous-to-Discrete Ranking Transfer). *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i, j) \in E$, $v_k = \|[S_c]_{:,k}\|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the discrete damage from fixed-normalization removal of edge k . Assume all single-edge removals are subcritical: $\sqrt{2} \max_k w_k < \varepsilon_{\text{crit}}$.*

(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \quad (8)$$

where $L_J \leq \|W\|_2^2 \|z^\|$ is the second-order curvature constant of $z^*(A)$, finite at the fixed point.*

(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J / (1 - \kappa)^2}. \quad (9)$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2} (v_{k_1} - v_{k_2}) \cdot (1 - \kappa)^2 / L_J$.

Proof. (a). We model deletion as fixed-normalization masking, $[\delta \hat{A}]_{ij} = [\delta \hat{A}]_{ji} = -w_k$ with the degree matrix D held fixed, matching the continuous edge-weight perturbation, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. (recompute-normalization adds an $O(d_i^{-1})$ incident-edge rescaling, negligible off low-degree nodes; agreement in section V-A.) For the remainder, the IFT gives $\partial z^* / \partial \text{vec}(A) = (I - J_z)^{-1} J_A$, and the dominant curvature term $(I - J_z)^{-1} (\partial J_z / \partial \text{vec}(A)) (I - J_z)^{-1} J_A$ carries two resolvents $(1 - \kappa)^{-1}$, two factors $\|W\|_2$, and the equilibrium norm $\|z^*\|$ (since $\partial J_z / \partial \text{vec}(A) \propto W$ and $J_A \propto z^* W^\top$), so $L_J \leq \|W\|_2^2 \|z^*\|$, finite since $\|z^*\| \leq \|X_{\text{proj}}\| / (1 - \|\hat{A}\|_2 \|W\|_2)$ (from $\|\phi(u)\| \leq \|u\|$ at the fixed point and the spectral normalization $\|\hat{A}\|_2 \|W\|_2 < 1$ of section II), giving $\|\partial^2 z^* / \partial \text{vec}(A)^2\|_2 \leq (1 - \kappa)^{-2} L_J$. The path meets finitely many ReLU regions (simultaneous crossings non-generic, measure zero); summing the second-order remainder over these regions, with the conservative IFT [30] on each, gives $|R_k| \leq L_J w_k^2 / (2(1 - \kappa)^2)$. **(b).** From (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J / (2(1 - \kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \geq w_{k_2}$, the first term is positive and (9) ensures the first-order gap dominates. $\square$

Empirically $|R_k|$ is $2\text{--}10\times$ smaller than $L_J w_k^2$; the constrained projection S_c turns the 0.31 vacuous bound into a tight one (1.00; section V-A) [31], [26]. The edge-weighted score $w_k v_k$ predicted by (a) reaches $\tau = +0.996$ against brute-force N-1 on full-graph Amazon Photo (10 seeds, section V-F), corroborating the first-order bridge at scale.

B. Generalization to Explicit GNNs

Observation 8 (Structural Sensitivity for K -Layer GNNs). *Let $Z_K = g_K \circ \dots \circ g_1(X, A)$ be a K -layer GNN where each layer g_l is differentiable with respect to both its input and A . Define the unrolled sensitivity matrix*

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)}, \quad (10)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$. Then:

(a) *The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \leq \sum_{l=1}^K \left(\prod_{k=l+1}^K \|J_z^{(k)}\|_2 \right) \|J_A^{(l)}\|_2. \quad (11)$$

(b) *The constrained construction S_c and per-node radius r_v (theorem 5) apply identically with S_K in place of S .*

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2 (1 - \kappa^K) / (1 - \kappa)$ converges to the IGNN bound as $K \rightarrow \infty$. Explicit GNNs inherit the full computational pipeline with empirical tightness 0.99–1.02 across the architecture suite (section V-F); the closed-form $\varepsilon_{\text{crit}}$ is available only for the contractive subclass.

IV. AEGIS CONSTRUCTION

The pipeline turns section III into a practical tool through matrix-free computation, sidestepping both the materialisation of $S \in \mathbb{R}^{Nd \times Nd}$ and the formation of the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta \hat{A}^*$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\text{crit}}$ and the regime characterisation of theorem 1 for IGNN-class operators. A dense path ($N \leq 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to $N=7,650$) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^K J_z^k b$ with depth K adaptive in κ (early stop at $\|J_z^k b\| < 10^{-6} \|b\|$); each Neumann term and $J_A \text{vec}(\delta A)$ are forward-mode JVPs in $O(Nd)$ time. P_c is applied implicitly, so neither S_c nor S is formed; the dense path ($N \leq 200$) materialises J_z, J_A and applies a direct linear solve. The rSVD [33] ($k=p=10$, $n_{\text{iter}}=2$) returns (u_1, σ_1, v_1) for the direction $\delta \hat{A}^* = \varepsilon \cdot \text{sym}(P_c v_1)$ (theorem 4); column norms give $\{v_{ij}\}$, and per-node radii follow theorem 5 via the node-restricted RMATVEC. The well-separated singular spectrum (gap $(\sigma_1 - \sigma_2) / \sigma_1 = 0.39\text{--}0.50$; fig. 2) is why one rSVD query matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $O(K \cdot Nd)$ time and $O(Nd)$ memory ($K \in [20, 50]$ for $\kappa < 0.8$); removing the $O((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section V-D).

Algorithm 1 AEGIS Vulnerability Analysis (matrix-free). Assumes the equilibrium z^* is pre-computed on the analysis (sub)graph. Dense fallback ($N \leq 200$): form $S = (I - J_z)^{-1} J_A$ and run a deterministic SVD in place of lines 2–11.

Require: $F, z^*, \hat{A}$, budget ε ; SVD rank k , oversampling p , power iters n_{iter} ; Neumann tol. τ ; edge set E ; (opt.) head h , logits Y , labels y .

Ensure: $\sigma_1(S_c), \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}}$.

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1:  $\kappa \leftarrow \|J_z\|_2$  via power iteration;  $K \leftarrow \lceil \log(1/\tau) / \log(1/\kappa) \rceil$ .
2:  $\text{MATVEC}(v) := \sum_{j=0}^K J_z^j \text{JVP}_F^{\hat{A}}(P_c v)|_{z^*}$ ;
    $\text{RMATVEC}(u) := P_c^\top \text{VJP}_F^{\hat{A}}(\sum_{j=0}^K (J_z^\top)^j u)|_{z^*}$ .
3: Draw  $\Omega \sim \mathcal{N}(0, I) \in \mathbb{R}^{|E| \times (k+p)}$ ;  $Y \leftarrow \text{MATVEC}(\Omega)$ 
   column-wise.
4: for  $j = 1, \dots, n_{\text{iter}}$  do
5:    $Y \leftarrow \text{QR}(\text{MATVEC}(\text{RMATVEC}(Y)))$ 
6: end for
7:  $Q \leftarrow \text{QR}(Y)$ ;  $B^\top \leftarrow \text{RMATVEC}(Q)$ ;  $(\tilde{U}, \Sigma, V^\top) \leftarrow \text{SVD}(B)$ ;  $U \leftarrow Q \tilde{U}$  {rSVD [33]}
8:  $\sigma_1 \leftarrow \Sigma_{1,1}$ ;  $\delta \hat{A}^* \leftarrow \varepsilon \cdot \text{sym}(P_c V_{:,1}) / \|\text{sym}(P_c V_{:,1})\|_2$ .
9:  $v_{ij} \leftarrow \|\text{MATVEC}(e_{ij})\|_2$  for each  $(i, j) \in E$  {per-edge spectrum}
10: if  $h, Y, y$  supplied then
11:    $r_v \leftarrow \min_{c \neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c) S_{c,v}\|_2$  for each  $v$ 
     {Prop. 5}
12: end if
13: if  $\|W\|_2$  extractable then  $\varepsilon_{\text{crit}} \leftarrow (1 - \kappa) / \|W\|_2$  {Thm. 1}
14: return  $(\sigma_1, \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$ .
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V. EXPERIMENTS

Setup. We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (PyTorch [34]). IGNN [22] uses $F(Z) = \text{ReLU}(\hat{A} Z W^\top + U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised [23] and Adam (lr 0.01); ContractiveGCN-PF mirrors this with 5 bus features and 2-head readout. The spectral-norm constraint costs $\sim 6\%$ Cora accuracy (77.5% vs. $\sim 83\%$). Datasets: Cora, Citeseer, Pubmed [35], Amazon Photo [36], WikiCS [37], IEEE 14/30/57/118 [38], [39]. Baselines: smoothing (Gaussian $\sigma=0.01$, 200 MC, Clopper-Pearson $\alpha=0.001$), Mettack with Meta-Self GCN surrogate, LODF as $\text{PTDF}_{ij} / (1 - \text{PTDF}_{jj})$ from the DC B -matrix [40, Ch. 11]. The 9 datasets and 7 architectures are cumulative: all seven architectures appear only in table V and fig. 7 (33 cells); other studies use IGNN, and power flow the four IEEE cases.

A. Cross-Domain Adversarial Analysis

Table I verifies the contractivity assumption (A3) on every benchmark ($\kappa < 1$) and reports the per-dataset closed-form $\varepsilon_{\text{crit}}$ alongside an attack advantage of $3.2\text{--}4.1\times$ over random perturbations. Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs centred on the highest-degree node and report *local vulnera-*

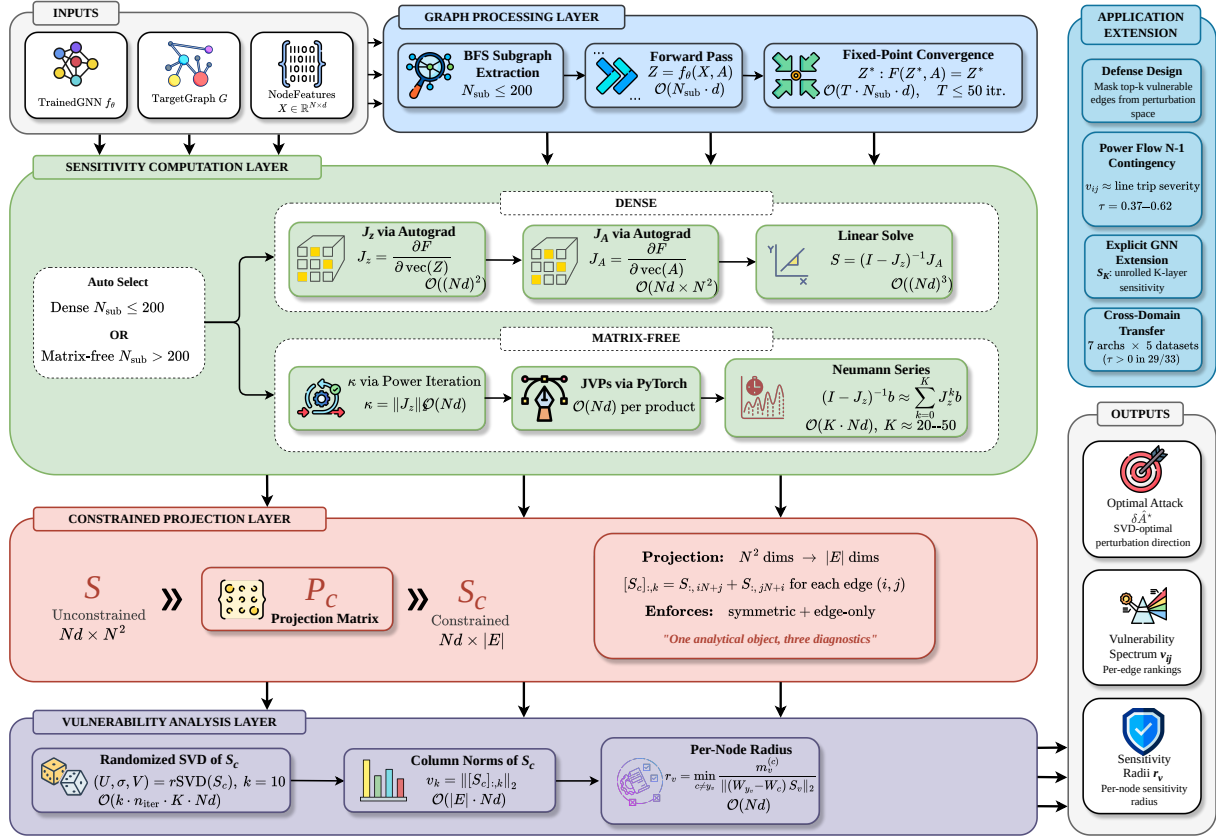


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \rightarrow |E|$ via P_c ; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJPs without materialising it, scaling to $N=7,650$.

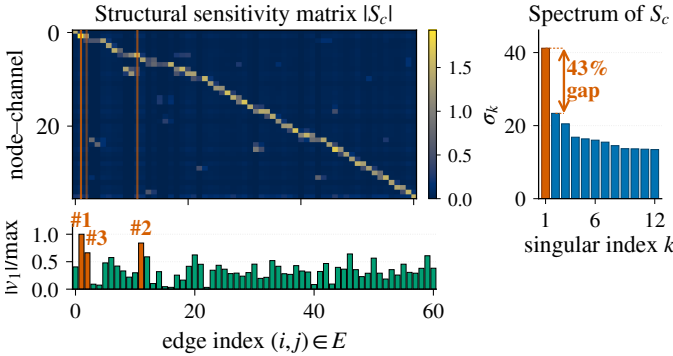


Fig. 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, $N=50$, $d=16$, $|E|=61$, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, $\sigma_1=41.2$, 43% gap to σ_2 (cross-benchmark $(\sigma_1-\sigma_2)/\sigma_1 \in [0.39, 0.50]$; section V-B).

bility; graph-level assessment uses the matrix-free pipeline of section V-D. All bounds use the operator-norm $\kappa = \|J_z\|_2$.

The envelope ratio is measured on equilibrium shift $\|\Delta z^*\|_F$ (table II), distinct from the prediction-flip (breach) rate of fig. 4 (flips stay $< 2\%$ at $\varepsilon \leq 0.10$, $< 6\%$ at $\varepsilon = 0.20$).

Heuristic and surrogate baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ is the theoretical critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes certified first-order-safe at budget $\varepsilon=0.05$ (i.e. corrected radius $r_v > 0.05$).

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	77.5 $\pm$ 1.7	.33 $\pm$.14	.66 $\pm$.15	3.6 $\pm$.6	83 $\pm$ 10
Citeseer	66.0 $\pm$ 0.7	.59 $\pm$.02	.41 $\pm$.02	4.1 $\pm$.5	94 $\pm$ 5
Pubmed	78.9 $\pm$ 0.7	.41 $\pm$.11	.59 $\pm$.11	3.2 $\pm$.4	76 $\pm$ 18
Amazon	94.8 $\pm$ 0.3	.14 $\pm$.02	.86 $\pm$.02	3.8 $\pm$.2	86 $\pm$ 8
WikiCS	77.9 $\pm$ 0.4	.34 $\pm$.04	.66 $\pm$.04	3.8 $\pm$.4	78 $\pm$ 4

TABLE II: First-order envelope ratio (actual / first-order-predicted equilibrium shift) with increasing ε (10 seeds, S_c -optimal constrained perturbation). Tight at $\varepsilon=0.01$ (≈ 1.00); under-predicts by up to 39% at $\varepsilon=0.20$.

Dataset	$\varepsilon=.01$	$\varepsilon=.05$	$\varepsilon=.10$	$\varepsilon=.20$
Cora	1.01 $\pm$.00	1.07 $\pm$.01	1.15 $\pm$.03	1.36 $\pm$.08
Citeseer	1.01 $\pm$.00	1.07 $\pm$.01	1.16 $\pm$.03	1.39 $\pm$.07
Pubmed	1.01 $\pm$.00	1.07 $\pm$.01	1.15 $\pm$.02	1.38 $\pm$.06
WikiCS	1.00 $\pm$.01	1.01 $\pm$.01	1.02 $\pm$.01	1.05 $\pm$.01
Amazon	1.00 $\pm$.00	1.01 $\pm$.00	1.03 $\pm$.01	1.07 $\pm$.01

$\varepsilon=0.01$ (+6–148% AtkAdv, Wilcoxon $p<0.001$) and inflicts 3–10 $\times$ more equilibrium damage than Mettack [4] at the early-warning regime $k \in \{1, \dots, 5\}$ (149/150 paired wins, $p<10^{-43}$); GR-BCD/PR-BCD [5] and the discrete-removal curves are in table IV and fig. 3.

TABLE III: Equilibrium damage at $\varepsilon=0.10$ across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70 \pm .79$	$2.51 \pm .48$	$3.05 \pm .70$	$0.97 \pm .25$
Citeseer	$4.63 \pm .71$	$2.97 \pm .42$	$3.35 \pm .57$	$1.01 \pm .15$
WikiCS	$2.10 \pm .22$	$0.67 \pm .11$	$1.93 \pm .20$	$0.53 \pm .09$

B. Four-Quadrant Attack Comparison

Radius vs. smoothing. Randomized smoothing [7] gives constant-per-node base radii (0.123 vs. AEGIS’s 0.078 at $\sigma=0.05$) and localized smoothing [9] needs 10^3 – 10^4 MC/node, both lacking edge structure; AEGIS radii are structurally informative ($r_v \approx 0.10$ dense, 0.01 at boundaries), a complementary point on the certificate-vs-diagnostic frontier.

Table III evaluates AEGIS against baselines spanning the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). The one-query SVD direction dominates: a 50-step equilibrium-shift PGD [41] using AEGIS’s own IFT gradients recovers only 72–92% of its damage (solver validation, not an independent baseline), and a classification-loss PGD inflicts 15–70% less while flipping predictions at comparable rates (0–1.8%). The decisive test is a *gradient-independent* adversary: a transfer direction from a separately trained surrogate IGNN (zero shared gradients) recovers only 99% of the one-query damage ($\cos=0.99$ alignment), and a 512-query black-box search just $44 \pm 4\%$. The direction is therefore model-intrinsic, not a circular gradient artifact: we term it the *maximally sensitive first-order direction*, a one-query guarantee that empirically matches iterative attackers.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20 \pm 0.05$ vs. AEGIS’s $+0.32 \pm 0.04$) and stay near zero elsewhere; AEGIS achieves 2–6 $\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Discrete edge removal. See fig. 3 for four sequential orderings.

Per-edge finite differences reproduce the S_c column-norm ranking ($\tau=0.999$); iterative S_c recomputation closes ≤ 3 pp of the static-vs-greedy gap at $k \times$ compute, so the static ranking is the headline.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a reliable first-order screening threshold (theorem 6). Figure 4 reports prediction-flip fractions across ε (bands are 95% CIs): Citeseer and WikiCS stay below 2% throughout; Cora reaches 7.6% (CI [2.2, 13.0]) at $\varepsilon=0.20$; Pubmed is the right-skewed outlier (median 7.8%, mean 10.3% at $\varepsilon=0.10$; 27.4%, CI [12.3, 42.5], at $\varepsilon=0.20$).

C. Comparison with Prior Structural Baselines

Table IV reports head-to-head numbers against GR-BCD and PR-BCD [5], AGNNCert [11], and LODF [40]. AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains.

TABLE IV: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper’s larger-budget variant, see text); AGNNCert [11] is a sound IBP certifier (semantics differ from AEGIS r_v); LODF [40] is the industry DC-PF screener.

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed $k=10$	0.356	0.350	+0.69
GR-BCD [5]	Cora $k=5$	0.643	1.207	+0.16
AGNNCert [†] [11]	Cora med. r_v	0.163	1.414	+0.08
LODF [40]	case118 P@10	0.81	≤ 0.20	+0.14

[†]AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (theorem 6). Per-dataset mean tightness ratio $r_{\text{cert}}/r_v \in [4.4, 15.0]$ (per-cell across 30 seeds [2.1, 39.0] $\times$); per-seed Kendall $\tau \in [-0.11, 0.24]$. **Decision rule:** if AGNNCert certifies node v safe at budget ρ , trust it; if $r_v < \rho$ and AGNNCert does *not* certify, v is first-order unsafe and warrants attention. The methods are complementary in this prescribed sense, not interchangeable.

GR-BCD [5] aligns with S_c on Pubmed ($\tau=+0.69$) but decorrelates on Cora ($\tau=+0.16$): AEGIS is a label-free one-pass diagnostic, not a budget-optimal attacker, so it is competitive rather than dominant under direct attack damage (table IV). AGNNCert [11] is complementary—a sound IBP certificate vs. first-order radii 4.4–15.0 $\times$ tighter that expose the dominant direction (decision rule: table IV). Power-grid LODF/PI comparisons are in section VI.

D. Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\max} \in [0.30, 0.99]$ on Cora (10 seeds): even when the spectral-normalisation cap is pushed to 0.99 (driving the theoretical $\varepsilon_{\text{crit}}$ down 230 $\times$), the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent grows only 1.17 $\rightarrow$ 1.80, *validating* $\varepsilon_{\text{crit}}$ as a closed-form sufficient safety boundary that holds with 2–4 $\times$ empirical margin across the sweep. Training converges throughout; the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\max} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 5 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at $N=200$ (per-edge $\tau=0.999$), and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (consistent with the trained-IGNN range $\kappa \in [0.14, 0.59]$ of table I). For larger N the rSVD error [33] is bounded by the spectral gap; Pubmed ($N=19,717$) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N=200$; we use $N=50$ as default (66 $\times$ faster). Subgraph-vs-full-graph agreement is high on dense grids (case14 $\tau=0.80$ –0.86).

Full-graph scale. A 50-node BFS covers only $\sim 1.8\%$ of a citation graph’s edges (Cora $\tau=0.16$), so at this scale we run AEGIS on the full graph, where its edge advantage over degree-proportional *amplifies* to 9.82 $\times$ (Citeseer) and 3.25 $\times$ (Cora) at $k=10$, vs. $\approx 1.1 \times$ on subgraphs.

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$

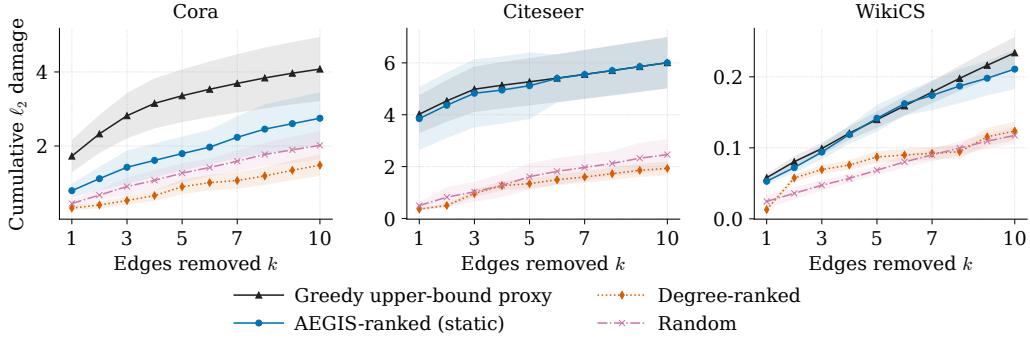


Fig. 3: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS matches the ground-truth-aware Greedy attacker on Citeseer/WikiCS and recovers 54–67% of Greedy’s damage on Cora at $k=5-10$ without any access to discrete ground truth; degree-ranked stays below random on Cora at every k .

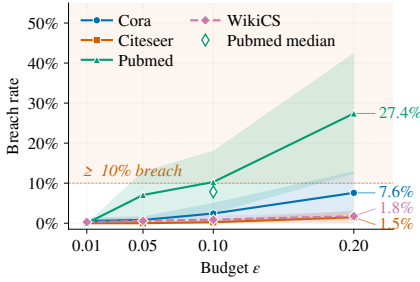


Fig. 4: Breach rate (% of nodes flipped under S_c -optimal perturbation) vs. ε (10 seeds; mean, 95% CI bands).

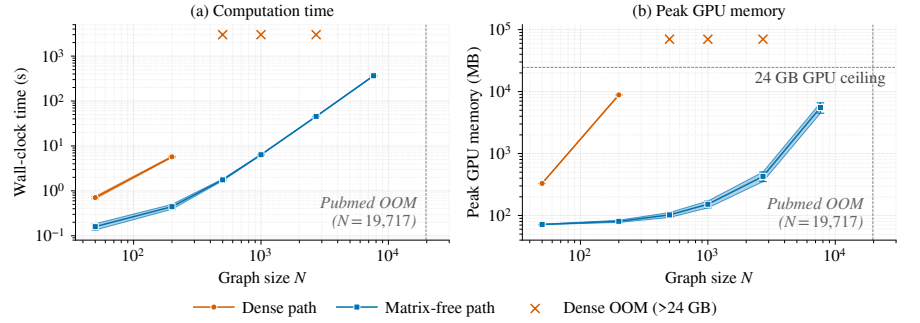


Fig. 5: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond $N=200$; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo ($N=7,650$) at 365 s/5.5 GB.

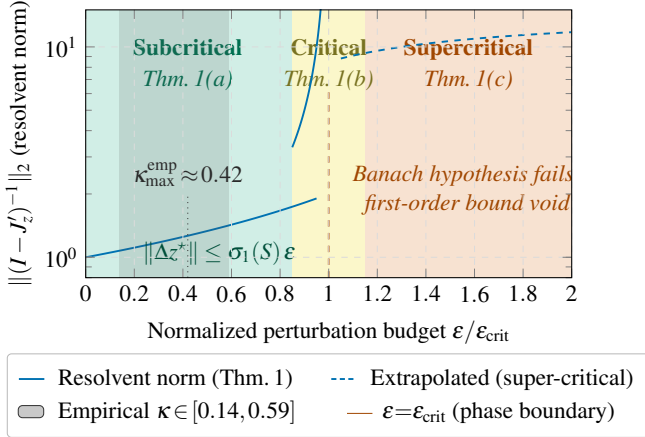


Fig. 6: Three-regime characterisation of theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \rightarrow \varepsilon_{\text{crit}}$.

traces the accuracy–robustness frontier ($r_v=0.147@72.1\%$ vs. $0.089@80.6\%$).

Defense-informed edge protection: gains survive adaptive attack. Masking the top- k edges by v_{ij} (Cora IGNN, $N=50$, 10 seeds) cuts $\sigma_1(S_c)$ -damage by $42\pm 8\%$ at $k=5$ and $61\pm 7\%$ at $k=10$, vs. $11\pm 6\%/18\pm 9\%$ for random masking (paired

Wilcoxon $p<0.002$). An *adaptive* attacker that recomputes S_c on the masked graph matches the non-adaptive attacker within 0.1 pp (10 seeds, $\varepsilon=0.10$): the 43% spectral gap of S_c (fig. 2) keeps the dominant direction robust, so the defense *does not erode under recomputation*—the failure mode sober-look critiques [16] flag.

F. Extension to Explicit GNNs

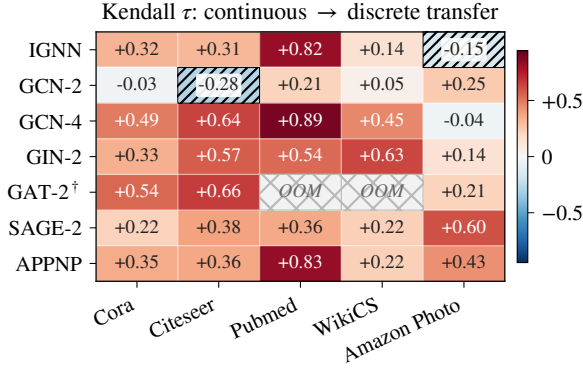
For K -layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT[†] [42], [43], [44], [45], [46]) we form $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ via finite differences and construct S_c from S_K identically (theorem 8). Table V reports the per-architecture characterisation on Cora: IGNN carries theorem 1; the explicit models receive the computational tool without the regime characterisation.

GAT[†] scope. Standard GAT uses A as a binary mask, so $\partial Z / \partial A_{ij} = 0$ on existing edges and S_c is undefined; our GAT[†] modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij} \alpha_{ij} W h_j$), restoring differentiability and yielding tightness 0.99, AtkAdv $4.4\times$, $\tau = +0.36$. Binary-mask architectures (hard attention, max/min aggregation, GATv2 [47]-style dynamic-attention masking) fall outside the framework.

Robust backbones. Under the $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite [13] ($\tau = +0.37 / +0.54$, Cora/Citeseer) and GNNGuard-lite [14] ($+0.10 / +0.53$) match or exceed IGNN; the cap is a precondition (without it $\kappa > 1$ voids theorem 1).

TABLE V: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ : Kendall vs. brute-force fixed-normalization single-edge removal (theorem 7). Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	1.01 $\pm$.00	7.6 $\pm$ 0.5 $\times$	+ .32 $\pm$.04	77.5 $\pm$ 1.7%
GCN-2	1.00 $\pm$.00	3.6 $\pm$ 0.6 $\times$	−.04 $\pm$.03	78.9 $\pm$ 0.9%
GCN-4	1.02 $\pm$.00	3.4 $\pm$ 0.4 $\times$	+ .49 $\pm$.02	78.3 $\pm$ 1.8%
GIN-2	1.01 $\pm$.00	2.6 $\pm$ 0.1 $\times$	+ .33 $\pm$.03	76.6 $\pm$ 1.9%
GAT [†]	1.01 $\pm$.01	2.1 $\pm$ 0.1 $\times$	+ .54 $\pm$.06	77.8 $\pm$ 1.2%
SAGE-2	1.00 $\pm$.00	1.9 $\pm$ 0.2 $\times$	+ .22 $\pm$.10	77.5 $\pm$ 1.5%
APPNP	1.02 $\pm$.00	3.4 $\pm$ 0.3 $\times$	+ .35 $\pm$.01	82.2$\pm$0.6%



[†] GAT-2: 2-layer attention; OOM on Pubmed and WikiCS.

Fig. 7: Continuous-to-discrete transfer τ across 7 architectures and 5 datasets (10 seeds, 330 runs). Cell values are mean Kendall τ ; buckets: strong ($\tau > 0.5$), moderate (0.3–0.5), weak (0–0.3), anti-correlated ($\tau < 0$, diagonal hatch); cross-hatch: GPU OOM. Deeper models (GCN-4, APPNP) and GAT[†] dominate.

Cross-dataset transfer. Figure 7 reports τ over 5 datasets and 7 architectures (330 runs); 29/33 cells positive (one-sided sign test $p < 10^{-5}$). Cold cells (GCN-2 on sparse citation graphs, IGNN on dense product graphs at 50 nodes) recover decisively at full-graph scale: on Amazon Photo ($N=7,650$, stratified top- v_{ij} ground-truth sampling) the edge-weighted ranking $A_{ij}v_{ij}$ predicted by theorem 7 reaches $\tau = +0.996 \pm 0.002$ (10 seeds) against brute-force fixed-normalization N-1 damage, a near-perfect rank match across seeds. On heterophilic WebKB [48] GCN-4 attains $\tau = +0.21$ – $+0.44$.

Theorem 7’s sufficient condition holds for **47–62%** of edge pairs across architectures; GCN-2’s 2-hop aggregates produce near-uniform first-order shifts and fall outside the sufficient regime, which the framework correctly flags via a low dataset-level τ (fig. 7).

VI. CASE STUDY: POWER FLOW CONTINGENCY

Beyond classification, AEGIS extends to AC power-flow contingency analysis: applied to a learned ContractiveGCN-PF surrogate, the S_c ranking recovers brute-force N-1 critical-line severity on IEEE case14–118 ($\tau = +0.37$ to $+0.62$, **P@10=0.66–0.81**), competitive with industry LODF screening. The ranking is label-free and one-query: v_{ij} comes purely from the surrogate’s learned IFT-resolvent $(I - J_z)^{-1}$, and its agreement with PTDF/LODF [40] is consistent with the

TABLE VI: AEGIS on IEEE power-flow benchmarks (10 seeds). Model quality: per-unit RMSE for $|V|$ and angle θ (std < 0.003). τ : Kendall correlation against brute-force N-1; P@10: top-10 precision.

Case	N	$ E $	Model (p.u.)		AEGIS	
			$ V $	θ	τ	P@10
case14	14	20	.007	.020	+ .42 $\pm$.19	.74 $\pm$.12
case30	30	41	.014	.024	+ .37 $\pm$.17	.68 $\pm$.06
case57	57	78	.033	.059	+ .67 $\pm$.09	.66 $\pm$.13
case118	118	179	.014	.076	+ .62 $\pm$.11	.81 $\pm$.10

topology-driven flow concentration those DC measures encode. Recent GNN-PF work targets fidelity [49], [50], [51], [3] and contingency screening [52]; AEGIS adds a label-free vulnerability-attribution layer over any such surrogate, without contingency labels or retraining.

Setup. ContractiveGCN-PF is an IGNN-class operator $F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ with spectral-norm-constrained W , trained on five IEEE test cases [38] with a 2-head readout $(\Delta|V|, \Delta\theta)$. Training data are 2,000 load samples per case (70–130% of nominal, *uniform load scaling only*, a narrow envelope that does not cover seasonal peaks, dispatch shifts, or renewable ramps) from PandaPower’s [39] AC Newton-Raphson solver; inputs are binary adjacency and 5 bus features (slack/PV indicators, P , Q , $|V|$). Ground truth: ℓ_2 voltage-angle deviation per AC contingency. We use full-graph dense analysis on case14–118. LODF’s native metric is DC line-flow change; we benchmark its rank-order against AC voltage-angle truth, where the comparison stays informative within LODF’s scope.

table VI reports the envelope ratio and Kendall τ between S_c rankings and brute-force N-1; fig. 8 visualises case14. The envelope is tight in the PF domain (≈ 1.00), and on case14–118 the ranking correlation $\tau = +0.37$ to $+0.62$ (P@10=0.66–0.81) identifies 7–8 of the top-10 critical lines. The matrix-free S_c pipeline scales to $N=7,650$ (fig. 5); richer PF surrogates trained on broader operating envelopes are the path to operational-grade screening.

Baselines. Against the AC voltage-angle truth, LODF (industry DC screening) attains $\tau = 0.44$ – 0.58 on the credible cases; AEGIS matches or leads it (0.62–0.67 on case57/118, Wilcoxon $p < 0.01$; overlapping on case14/30), and on LODF’s fairer thermal-overload target reaches P@10=0.60 (case57)—competitive, not dominant. *Binary adjacency beats admittance-weighting* (P@10=0.81 vs. 0.27, case118): a single-line trip removes the line regardless of impedance.¹

VII. RELATED WORK

We position AEGIS against three threads; each supplies at most one of the three diagnostics AEGIS produces from one S_c object via a single matrix-free pipeline.

¹Other baselines (case57/118): Ejebe–Wollenberg PI [53] $\tau = +0.335/+0.101$ (P@10=0.50/0.30); standalone PTDF without outage correction is anti-correlated/near-zero ($\tau = -0.14/+0.06$), confirming outage redistribution is the operative effect. Runtime: LODF < 0.13 s, N-1 0.1–2 s, AEGIS 2–23 s.

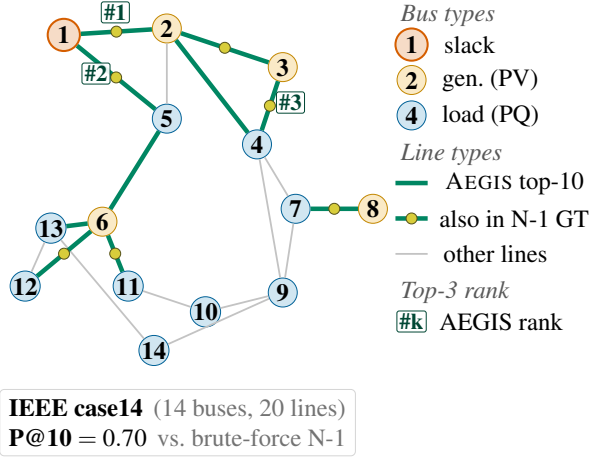


Fig. 8: AEGIS vulnerability ranking on the IEEE 14-bus benchmark (10-seed mean; $P@10 = 0.70$ for this run, consistent with the 0.74 ± 0.12 in table VI). Bus markers encode physical bus type (slack / generator / load); green edges are the top-10 by v_{ij} with rank tags (#1–#3); yellow midpoint dots mark edges that also appear in the brute-force N-1 top-10 (7 of 10). Top-ranked edges concentrate on the high-flow inter-area tie lines (1–2, 1–5, 7–8), recovering the single-contingency severity pattern from one matrix-free SVD pass.

Structural attacks. Nettek [1], Mettack [4], GR-BCD/PR-BCD [5] and variants [6] optimise surrogate gradients to flip predictions but provide no analytic maximally sensitive first-order direction and no per-node radius. Sober-look studies [16], [17] flag evaluation pitfalls; we follow their adaptive-attack discipline in section V-E (paired Wilcoxon against random masking; an adaptive-attacker column in defense ablation).

Certified robustness and defense. Convex relaxations [12], smoothing [7], [8], [9], collective certificates [10], and IBP [11] certify per-node robustness but do not identify vulnerability-driving edges. Empirical defenses [13], [15], [14] harden models without surfacing per-edge sensitivity. r_v is a first-order threshold, not a sound certificate (theorem 6); the methods are complementary under the prescribed decision rule of table IV.

Implicit networks and influence functions. DEQs [20], IGNN [22], monotone equilibria [54], and well-posedness analyses [28] address *input* sensitivity $\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$ via the resolvent $(I - J_z)^{-1} J_A$ and the unrolled Jacobian S_K . The construction descends from the Koh–Liang influence-function lineage [18] and from implicit-differentiation hyperparameter optimisation [27], [19], but those lines apply IFT to feature- or weight-perturbations of a loss; AEGIS applies it to a *structural-parameter* perturbation of the equilibrium state, projects onto the edge subspace via P_c , and obtains the SVD-optimal direction in a single matrix-free pass rather than via iterative reverse-mode. GNN explainers [55] target prediction fidelity (which subgraph rationalises an output), not adversarial vulnerability; classical matrix perturbation theory [31], [26] supplies the pseudospectral toolkit.

VIII. CONCLUSION

AEGIS produces three adversarial diagnostics (per-edge rankings, the SVD-optimal attack direction, per-node first-order

sensitivity radii) from one analytical object, the constrained sensitivity matrix S_c , computed matrix-free for any GNN with continuous edge-weight-modulated message passing, with a regime characterisation (theorem 1) that holds as a sufficient safety boundary for contractive implicit models.

Limitations. (i) Radii r_v are first-order thresholds, not probabilistic certificates (theorem 6); standard GAT and binary-mask architectures fall outside the framework (section V-F). (ii) Full-graph matrix-free analysis is the recommended graph-level path: AtkAdv amplifies ($3.3 \times / 9.8 \times$ on Cora/Citeseer) and the edge-weighted ranking $A_{ij}v_{ij}$ achieves $\tau = +0.996$ on Amazon Photo ($N=7,650$; section V-F). (iii) Opting into the formal $\varepsilon_{\text{crit}}$ track accepts a $\sim 6\%$ accuracy cost, on par with certified-robustness trade-offs. (iv) Operator-grade power-flow deployment (admittance-weighted edges, broader load envelopes) is a natural follow-up. (v) Insertion attacks (Nettack-class) are deliberately scoped out; the S_c construction extends once the edge basis is enlarged to $\bar{E} \supseteq E$ (section II).

Disclosure protocol. (i) A 90-day coordinated-notification window before public release, addressed to the maintainers of OGB graph-robustness benchmarks and to the GR-BCD / Mettack / AGNNCert authors. (ii) Attack-direction code (Algorithm 1 steps 8–9) gated behind institutional-affiliation review mediated by an institutional research-ethics office (named in the camera-ready). (iii) The diagnostic-only path (r_v and v_{ij} scores; no SVD reconstruction) released unconditionally, since it cannot directly synthesise a perturbation. Power-grid artefacts use only public IEEE benchmarks [38].

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